

Amanpreet Singh

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SUMMARY

Machine Learning Engineer & Researcher with a background in statistics and 1.5 years of experience in end-to-end software development. Created and maintain repspat, an open-source Python package for nonparametric repeated spatial pattern detection, and delivered a workshop on it at the 2026 Bioconductor (BioC) conference. Peer-reviewed publication in deep learning. Hands-on across statistical modelling, deep learning (PyTorch), LLM and agentic systems, and applied ML on financial data.

RESEARCH

repspat | [GitHub](#)

Feb 2026 – Present

- Built an open-source Python package that finds repeated spatial patterns in tissue data, regions that sit apart but share similar cell profiles.
- Implemented a statistical significance test to validate detected patterns, and ensured compatibility with standard research tools for straightforward adoption.

Attention-Based Feature Selection for High-Dimensional Clustering | M.Sc. Thesis, McMaster University

Sep 2025 – Present

- Developing a method that performs clustering and feature selection jointly, optimized in a stage-wise manner to handle high-dimensional data.
- Applying multi-head self-attention as a feature-gating mechanism, so the model learns which features matter across the levels of a hierarchical clustering; manuscript in preparation.

Publications

- *Audio Classification of Emergency Vehicle Sirens Using Recurrent Neural Network Architectures* — [Springer](#), Oct 2023.

PROFESSIONAL EXPERIENCE

Teaching Assistant, Data Science | McMaster University | Hamilton, ON

Sep 2025 – Present

- Assisted in teaching core data science concepts such as statistical modelling, machine learning, and data visualization, guiding 100+ undergraduate students through labs and practical exercises.
- Supported student learning by grading assignments, holding office hours, and mentoring one-on-one on Python, R, and data analysis techniques.

Systems Engineer | Shell India Markets Pvt. Ltd. | Bengaluru, India

Aug 2023 – Dec 2024

- Owned the joiner-mover-leaver module on ServiceNow, a microservice-based component that manages employee onboarding, role changes, and offboarding, handling 500+ lifecycle transitions each year.
- Developed automation pipelines and integrated REST APIs to streamline workflows and reduce manual effort, using client- and server-side JavaScript.
- Conducted impact analysis on proposed changes ahead of each release, ensuring smooth rollouts and saving roughly two weeks of developer time per cycle.
- Analyzed ServiceNow Incident Management data to identify trends and root causes, reducing incident-resolution effort by 20% and improving SLA compliance.

EDUCATION

M.Sc., Computational Science and Engineering | McMaster University, Hamilton, ON

Dec 2026

- Funded graduate research student. Focus: nonparametric methods for repeated spatial clustering and high-dimensional statistical inference.
- CGPA 3.93/4.00.

B.Tech., Computer Science and Business Systems | NMIMS University, Mumbai, India

May 2023

- Foundational coursework in CS, statistics, and business systems; authored deep learning publication.
- CGPA 3.70/4.00.

PROJECTS

AI Market & Competitor Analysis Tool | [GitHub](#)

Jan 2026

- Developed a multi-stage LLM pipeline (Python, Streamlit, DeepSeek-V3.2 via HuggingFace Inference API) that decomposes a company query into targeted web searches, applies retrieval-augmented generation, and synthesizes a SWOT analysis into a streamed business-intelligence report.
- Turned unstructured web search results into structured, queryable insights using MiniLM embeddings, DistilBERT sentiment, BERT-NER entity extraction, and KMeans clustering.

RAG Pipeline for Context-Aware Q&A | [GitHub](#)

Dec 2025

- Created a Retrieval-Augmented Generation module (Python, LangChain, HuggingFace Inference API) that retrieves relevant knowledge from Wikipedia and generates context-aware answers with Meta-LLaMA 3.1 Instruct.
- Improved retrieval speed and accuracy with semantic chunking, near-duplicate filtering, and a persistent Chroma vector store (all-MiniLM-L6-v2 embeddings).

Cost-Optimized Credit Risk Modelling | [GitHub](#)

Nov 2025

- Designed and evaluated a loan-default prediction model within a cost-sensitive framework, incorporating the asymmetric costs of lending decisions to prioritize catching defaulters (0.86 recall on the default class) over raw accuracy.

End-to-End ML Pipeline (ZenML, MLflow) | [GitHub](#)

Jun 2025

- Engineered a modular, reproducible ML pipeline (ZenML, MLflow) for house-price prediction, automating data ingestion, feature engineering, model training, evaluation, and deployment with experiment tracking.

TECHNICAL SKILLS

Languages Python, SQL, R, JavaScript, Bash/Shell

ML & Deep Learning PyTorch, TensorFlow/Keras, scikit-learn, HuggingFace Transformers, NumPy, Pandas, SciPy, statsmodels

LLM & Gen AI RAG, agentic pipelines, LangChain, Chroma vector store, embeddings (MiniLM/Sentence Transformers), sentiment analysis (DistilBERT), NER (BERT), HuggingFace Inference API, LLaMA, DeepSeek, prompt engineering

MLOps & Infrastructure ZenML, MLflow, Docker, REST API design, ETL pipelines, experiment tracking, model deployment

Data & Cloud data warehousing, SSMS, Azure (AI-900), AWS, Tableau/Power BI